

SL10 / L2

Analisi dinamica

Traccia:

Esercizio Analisi dinamica Configurare la macchina virtuale per l'analisi dinamica (il malware sarà effettivamente eseguito). Con riferimento al file eseguibile contenuto nella cartella «Esercizio_Pratico_U3_W2_L2» presente sul desktop della vostra macchina virtuale dedicata all'analisi dei malware, rispondere ai seguenti quesiti:

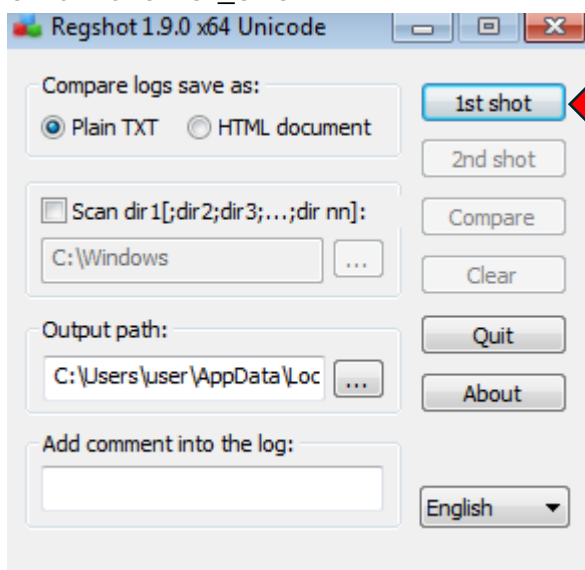
- Identificare eventuali azioni del malware sul file system utilizzando Process Monitor (procmon)
- Identificare eventuali azioni del malware su processi e thread utilizzando Process Monitor
- Modifiche del registro dopo il malware (le differenze)
- Provare a profilare il malware in base alla correlazione tra «operation» e Path

Tool Utilizzati

- 1) Regshot
- 2) Process Monitor (procmon)
- 3) Process Explorer
- 4) ApateDNS

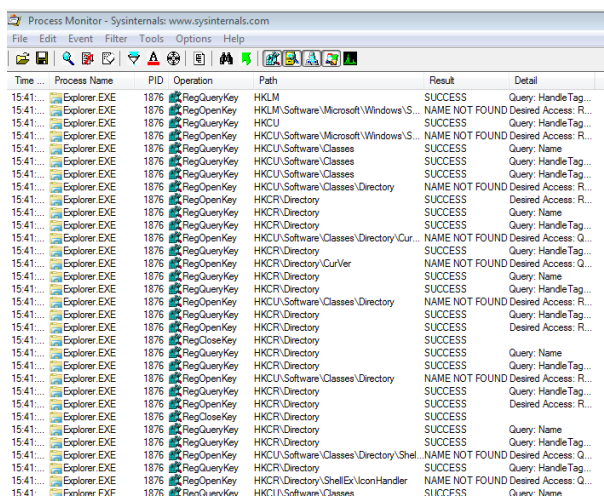
Preparazione

- 1) Creazione Macchina Virtuale Windows 7
- 2) Isolamento della macchina
- 3) Configurazione dei tool
 - a. Cominciamo avviando Regshot per salvare un'immagine dei registri prima dell'avvio del malware. Il risultato è stato salvato in un file chiamato 1st_shot.



Cliccare su **1st shot** per eseguire la prima scansione

- b. Dopo di che ho avviato il tool Process Monitor per metterlo in “ascolto” sul sistema



Avvio del Malware

- 1) Avviato il Malware ci troviamo davanti questa schermata



- 2) Per tracciare le attività del malware abbiamo bisogno del nome del processo. Possiamo ottenere il nome grazie al tool **Process Explor**.

6AdwCleaner.exe	< 0.01	29.956 K	18.620 K	1596 AdwareBooC
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Ora sappiamo che il nome del processo è **6AdwCleaner.exe**

- 3) Dopo aver lasciato per alcuni momenti il malware agire, lo chiudiamo così da poter analizzare i suoi movimenti senza ulteriori modifiche.

Analisi Registro

- 1) Avviamo il secondo shot con Regshot (salvato in un file chiamato 2st_shot) e clicchiamo su “**compare**” per analizzare le differenze

```
~res-x64 - Blocco note
File Modifica Formato Visualizza ?
Datetime: 2024/7/30 13:44:53 , 2024/7/30 13:51:38
Computer: USER-PC , USER-PC
Username: user , user

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Keys added: 6
HKLM\SOFTWARE\Microsoft\SystemCertificates\AuthRoot\Certificates\4EB6D578499B1CCF5F581EAD56BE3D986744A5E5
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS
HKLM\SOFTWARE\Wow6432Node\Microsoft\SystemCertificates\AuthRoot\Certificates\4EB6D578499B1CCF5F581EAD56BE3D986744A5E5
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\SysInternals\Process Explorer\ProcessComments
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\AdwCleaner

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Values added: 25
HKLM\SOFTWARE\Microsoft\SystemCertificates\AuthRoot\Certificates\4EB6D578499B1CCF5F581EAD56BE3D986744A5E5\Blob: 19 00 00 00 01 00 00 00 10 0
00 30 82 04 D3 30 82 03 BB A0 03 02 01 02 02 10 18 DA D1 9E 26 7D E8 BB 4A 21 58 CD CC 6B 3B 4A 30 0D 06 09 2A 86 48 86 F7 0D 01 01 05 05 0
0B 13 16 56 65 72 69 53 69 67 6E 20 54 72 75 73 74 20 4E 65 74 77 6F 72 6B 31 3A 30 38 06 03 55 04 0B 13 31 28 63 29 20 32 30 30 36 20 56 65
C 92 67 18 E5 F4 06 04 EF 90 B9 E4 00 E4 DD 3A B5 19 FF 02 BA F4 3C EE E0 8B EB 37 8B EC F4 D7 AC F2 F6 F0 3D AF DD 75 91 33 19 1D 1C 40 CB 7
BD 77 F6 A5 79 22 38 EC C4 A7 A0 78 12 AD 62 0E 45 70 64 C5 E7 97 66 2D 98 09 7E 5F AF D6 CC 28 65 F2 01 AA 08 1A 47 DE F9 F9 7C 92 5A 08 69
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\EnableFileTracing: 0x00000000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\EnableConsoleTracing: 0x00000000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\FileTracingMask: 0xFFFF0000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\ConsoleTracingMask: 0xFFFF0000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\MaxFileSize: 0x00100000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\FileDirectory: "%windir%\tracing"
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\EnableFileTracing: 0x00000000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\EnableConsoleTracing: 0x00000000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\FileTracingMask: 0xFFFF0000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\ConsoleTracingMask: 0xFFFF0000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\MaxFileSize: 0x00100000
HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\FileDirectory: "%windir%\tracing"
HKLM\SOFTWARE\Wow6432Node\Microsoft\SystemCertificates\AuthRoot\Certificates\4EB6D578499B1CCF5F581EAD56BE3D986744A5E5\Blob: 19 00 00 00 01 0
00 D7 04 00 00 30 82 04 D3 30 82 03 BB A0 03 02 01 02 02 10 18 DA D1 9E 26 7D E8 BB 4A 21 58 CD CC 6B 3B 4A 30 0D 06 09 2A 86 48 86 F7 0D 01
06 03 55 04 0B 13 16 56 65 72 69 53 69 67 6E 20 54 72 75 73 74 20 4E 65 74 77 6F 72 6B 31 3A 30 38 06 03 55 04 0B 13 31 28 63 29 20 32 30 30
2 25 F2 AF 1C 92 67 18 E5 F4 06 04 EF 90 B9 E4 00 E4 DD 3A B5 19 FF 02 BA F4 3C EE E0 8B EB 37 8B EC F4 D7 AC F2 F6 F0 3D AF DD 75 91 33 19 1
EA DC 99 2D BD 77 F6 A5 79 22 38 EC C4 A7 A0 78 12 AD 62 0E 45 70 64 C5 E7 97 66 2D 98 09 7E 5F AF D6 CC 28 65 F2 01 AA 08 1A 47 DE F9 F9 7C
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\Microsoft\Windows\CurrentVersion\Run\AdwCleaner: ""C:\Users\user\AppData\Local\6adw
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\SysInternals\Process Explorer\OriginalPath: "C:\Users\user\Desktop\ProcessExplorer\
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\SysInternals\Process Monitor\DeviceNameMap: ""
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\SysInternals\Process Monitor\FilterDialog: 2C 00 00 00 00 00 00 01 00 00 00 FF
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\SysInternals\Process Monitor\FilterDialogColumns: 64 00 00 00 64 00 00 00 64 00 0
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\Classes\Local Settings\MuiCache\4\F06864B\@C:\windows\system32\wusa.exe",-102: "P
HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Classes\Local Settings\MuiCache\4\F06864B\@C:\windows\system32\wusa.exe",-102: "pacchetto

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Values modified: 32
HKLM\COMPONENTS\ servicingStackVersions\6.1.7601.17514 (win7sp1_rtm.101119-1850): "2024/7/30:13:27:42.211 6.1.7601.17514 (win7sp1_rtm.101119-1
HKLM\COMPONENTS\ servicingStackVersions\6.1.7601.17514 (win7sp1_rtm.101119-1850): "2024/7/30:13:47:40.971 6.1.7601.17514 (win7sp1_rtm.101119-1
HKLM\SOFTWARE\Microsoft\SystemCertificates\AuthRoot\Certificates\4EB6D578499B1CCF5F581EAD56BE3D986744A5E5\Blob: 19 00 00 00 01 00 00 00 10 0
06 03 55 04 0B 13 02 55 34 31 17 30 15 06 03 55 04 07 13 0E 53 61 6C 74 20 4C 61 6B 65 20 43 69 74 79 31 1E 30 1C 06 03 55 04 0A 13 15 54 68
```

- 2) Notiamo che il malware ha aggiunto tre chiavi nei seguenti path:

- Chiave: **6AdwCleaner_RASAPI32**

Path:

HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32

- Chiave: **6AdwCleaner_RASMANCS**

Path:

HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS

- Chiave: **AdwCleaner**

Path: **HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\AdwCleaner**

- 3) Per ognuna di queste chiavi ha aggiunto una serie di valori:

- **Raspi32**

HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\EnableFileTracing: 0x00000000

HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\EnableConsoleTracing: 0x00000000

HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\FileTracingMask: 0xFFFF0000

HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\ConsoleTracingMask: 0xFFFF0000
 HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\MaxFileSize: 0x00100000
 HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASAPI32\FileDirectory: "%windir%\tracing"

- **Rasmancs**

HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\EnableFileTracing: 0x00000000
 HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\EnableConsoleTracing: 0x00000000
 HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\FileTracingMask: 0xFFFF0000
 HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\ConsoleTracingMask: 0xFFFF0000
 HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\MaxFileSize: 0x00100000
 HKLM\SOFTWARE\Microsoft\Tracing\6AdwCleaner_RASMANCS\FileDirectory: "%windir%\tracing"

- **AdwCleaner**

HKU\S-1-5-21-3771313050-58705377-3452663501-1001\Software\AdwCleaner\id: "0"

Analisi File System

Tramite Process Monitor notiamo che il malware ha fatto una serie di operazioni sul file system, andando a creare e operare con vari file.

Time ...	Process Name	PID	Operation	Path	Result	Detail
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Prefetch\6ADWCLEANER.EXE-A4D6D516.pf	NAME N...	Desired Access: Generic Read, Disposition: Open, Options: Sy...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Users\user\AppData\Local	SUCCESS	Desired Access: Execute/Traverse, Synchronize, Disposi...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\System32\mscoree.dll	SUCCESS	Desired Access: Read Attributes, Disposition: Open, Options: O...
15:50...	6AdwCleaner.exe	2672	QueryBasicInfor...	C:\Windows\System32\mscoree.dll	SUCCESS	CreationTime: 21/11/2010 05:23:48, LastAccessTime: 21/11/...
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\System32\mscoree.dll	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\System32\mscoree.dll	SUCCESS	Desired Access: Read Data/List Directory, Execute/Traverse, ...
15:50...	6AdwCleaner.exe	2672	CreateFileMappi...	C:\Windows\System32\mscoree.dll	FILE LOC...	SyncType: SyncTypeCreateSection, PageProtection:
15:50...	6AdwCleaner.exe	2672	CreateFileMappi...	C:\Windows\System32\mscoree.dll	SUCCESS	SyncType: SyncTypeOther
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\System32\mscoree.dll	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\System32\sechost.dll	SUCCESS	Desired Access: Read Attributes, Disposition: Open, Options: O...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\System32\sechost.dll	SUCCESS	CreationTime: 14/07/2009 01:20:52, LastAccessTime: 14/07/...
15:50...	6AdwCleaner.exe	2672	QueryBasicInfor...	C:\Windows\System32\sechost.dll	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\System32\sechost.dll	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\System32\sechost.dll	SUCCESS	Desired Access: Read Data/List Directory, Execute/Traverse, ...
15:50...	6AdwCleaner.exe	2672	CreateFileMappi...	C:\Windows\System32\sechost.dll	FILE LOC...	SyncType: SyncTypeCreateSection, PageProtection:
15:50...	6AdwCleaner.exe	2672	CreateFileMappi...	C:\Windows\System32\sechost.dll	SUCCESS	SyncType: SyncTypeOther
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\System32\sechost.dll	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\System32\MSCOREE.DLL.local	NAME N...	Desired Access: Read Attributes, Disposition: Open, Options: O...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319	SUCCESS	Desired Access: Read Data/List Directory, Synchronize, Dispos...
15:50...	6AdwCleaner.exe	2672	QueryDirectory	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	Filter: mscoree.dll, 1: mscoree.dll
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	Desired Access: Read Data/List Directory, Synchronize, Dispos...
15:50...	6AdwCleaner.exe	2672	QueryDirectory	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	Filter: mscoree.dll, 1: mscoree.dll
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	Desired Access: Read Attributes, Disposition: Open, Options: O...
15:50...	6AdwCleaner.exe	2672	QueryBasicInfor...	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	CreationTime: 18/03/2010 14:27:14, LastAccessTime: 30/07/...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	Desired Access: Read Data/List Directory, Execute/Traverse, ...
15:50...	6AdwCleaner.exe	2672	CreateFileMappi...	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	FILE LOC...	SyncType: SyncTypeCreateSection, PageProtection:
15:50...	6AdwCleaner.exe	2672	CreateFileMappi...	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	SyncType: SyncTypeOther
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\Microsoft.NET\Framework64\v4.0.30319\mscoree.dll	SUCCESS	
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\System32\MSCOREE.DLL.local	NAME N...	Desired Access: Read Attributes, Disposition: Open, Options: O...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64	SUCCESS	Desired Access: Read Data/List Directory, Synchronize, Dispos...
15:50...	6AdwCleaner.exe	2672	QueryDirectory	C:\Windows\Microsoft.NET\Framework64	SUCCESS	Filter: *, 1: ...
15:50...	6AdwCleaner.exe	2672	QueryDirectory	C:\Windows\Microsoft.NET\Framework64	SUCCESS	0: ..., 1: 1040, 2: sbcomp10.dll, 3: sbcomp20_mscorwks.dll, 4: sb...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64\v2.0.50727clr.dll	NAME N...	Desired Access: Read Attributes, Read Control, Synchronize, D...
15:50...	6AdwCleaner.exe	2672	CreateFile	C:\Windows\Microsoft.NET\Framework64\v2.0.50727mscorwks.dll	SUCCESS	Desired Access: Read Attributes, Read Control, Synchronize, D...
15:50...	6AdwCleaner.exe	2672	CloseFile	C:\Windows\Microsoft.NET\Framework64\v2.0.50727mscorwks.dll	SUCCESS	